

EMBU, Kenya

WMO: 637200

Lat: 0.5007S

Lon: 37.4591E

Elev: 1494

StdP: 84.62

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	13.8	6.2	7.0	20.6	7.7	7.8	19.9	4.7	18.3	4.2	18.9	0.9	0	0.264

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.0	28.8	17.4	27.9	17.3	27.1	17.4	19.6	24.7	19.2	24.3	18.9	24.0	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.5	22.0	17.4	15.0	21.5	17.1	14.7	21.2	62.9	24.8	61.4	24.5	60.4	24.2	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 6.7	5.4	4.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	8.9	20.6	1.3	0.7	8.0	21.1	7.2	21.6	6.4	22.0	5.5	22.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	20.6	21.5	21.7	21.0	20.2	18.7	17.9	18.4	19.8	20.7	20.1	19.9
	DBStd	1.66	0.95	1.15	1.46	1.17	0.87	1.20	1.37	1.45	1.49	1.27	0.96	0.88
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	60	0	0	0	0	1	10	25	18	5	1	1	1
	CDD10.0	3659	327	323	362	331	318	261	246	259	292	332	303	306
	CDD18.3	678	69	89	104	81	59	21	12	19	48	74	54	48
	CDH23.3	2598	359	580	643	223	74	14	19	41	189	256	92	107
CDH26.7	262	18	84	117	18	1	0	0	1	11	13	1	0	
Wind	WSAvg	1.6	2.6	2.7	2.0	1.5	1.1	0.9	0.9	1.1	1.4	1.6	1.7	2.0
Precipitation	PrecAvg	1173	30	27	106	283	167	29	29	35	30	163	233	53
	PrecMax	1928	144	155	224	563	688	131	180	105	92	620	605	218
	PrecMin	637	0	0	0	63	15	1	0	0	0	1	55	0
	PrecStd	335	40	40	65	117	116	26	29	24	24	132	119	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	29.7	30.0	28.5	26.0	24.6	25.1	26.0	27.9	28.0	26.2	26.3
		MCWB	17.0	17.0	17.7	17.9	18.6	17.6	16.6	16.6	17.2	17.8	18.4	17.7
	2%	DB	27.1	28.5	29.0	27.0	25.0	23.5	23.7	24.3	26.5	26.9	25.2	25.4
		MCWB	17.3	16.9	17.5	18.0	18.4	17.1	16.2	16.2	16.5	17.7	18.3	17.7
	5%	DB	26.2	27.5	28.1	25.7	24.1	22.8	22.6	23.1	25.4	25.9	24.4	24.6
		MCWB	17.3	17.1	17.4	18.2	18.1	16.8	16.0	16.0	16.2	17.3	18.2	17.7
	10%	DB	25.4	26.7	27.1	24.6	23.4	21.9	21.3	22.0	24.2	24.8	23.5	23.9
		MCWB	17.3	17.2	17.5	18.2	17.9	16.7	15.7	15.7	16.1	17.2	18.1	17.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	19.3	19.7	20.0	19.6	18.6	17.6	17.7	19.0	19.3	19.6	19.7
		MCDB	25.1	26.1	26.0	24.9	24.1	22.8	23.1	23.9	26.0	25.2	24.3	24.1
	2%	WB	18.9	18.6	19.1	19.4	19.0	17.9	16.8	17.0	17.5	18.6	19.0	19.1
		MCDB	24.4	25.7	25.6	24.1	23.6	22.3	21.9	22.6	24.1	24.4	23.6	23.7
	5%	WB	18.5	18.2	18.7	19.0	18.6	17.4	16.4	16.4	16.9	18.2	18.6	18.7
		MCDB	24.1	25.3	25.1	23.6	23.0	21.4	21.1	21.7	23.4	23.8	23.2	23.2
	10%	WB	18.1	17.7	18.3	18.6	18.2	17.0	16.0	15.9	16.4	17.8	18.3	18.2
		MCDB	23.8	24.9	24.5	23.1	22.4	20.9	20.4	20.9	22.5	23.1	22.7	22.6
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.8	11.0	8.3	7.2	6.7	7.0	7.5	9.5	9.1	7.7	8.9
		MCDBR	12.2	12.8	12.8	9.5	8.1	8.2	9.1	9.5	11.6	10.6	8.8	10.0
		MCWBR	6.0	5.2	4.8	3.5	3.2	3.3	3.5	3.5	4.0	3.6	3.8	5.1
	5% WB	MCDBR	10.3	11.4	10.3	8.3	7.6	7.2	7.9	8.4	10.2	9.0	8.1	8.9
		MCWBR	5.4	5.2	4.4	3.7	3.3	3.3	3.5	3.3	3.9	3.7	4.0	5.0
Clear-Sky Solar Irradiance	taub	0.367	0.387	0.381	0.341	0.336	0.389	0.402	0.376	0.354	0.355	0.356	0.366	
	taud	2.464	2.388	2.435	2.599	2.606	2.406	2.343	2.414	2.476	2.501	2.522	2.485	
	Ebn at Noon	963	944	942	959	940	875	869	912	953	964	967	959	
	Edh at Noon	116	127	121	99	95	114	123	118	114	112	109	113	
All-Sky Solar Radiation	RadAvg	6.24	6.56	6.34	5.65	5.34	4.60	4.26	4.54	5.66	5.81	5.40	5.69	
	RadStd	0.46	0.33	0.47	0.32	0.19	0.33	0.44	0.31	0.34	0.25	0.29	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.47	+0.69	N/A	+0.57	+0.39	+0.16	N/A	N/A	N/A	N/A

Nomenclature: See separate page